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FAS
APM1110

Question, Solution, Answer, formulas

$$P(x, y) = k(2x + y)$$

$$\mu = np$$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(x, y) = \binom{n}{x} p^x (1-p)^{n-x}$$

x	
1	3/24 = 0.1250
2	5/24 = 0.2083
3	7/24 = 0.2917
4	9/24 = 0.3750

① $x = 1, 2, 3, 4$

⑤ $k(2(1)+1) + k(2(2)+1) + k(2(3)+1) + k(2(4)+1) = 1$

$2k = 1$

$\frac{1}{24} = 0.0417$

$$E[X] = 1(0.125) + 2(0.2083) + 3(0.2917) + 4(0.3750)$$

$= 2.9167$

$$E[X^2] = 1^2(0.125) + 2^2(0.2083) + 3^2(0.2917) + 4^2(0.3750)$$

$= 9.583$

⑥ $P(X \geq 3)$

$= P(X=3) + P(X=4)$

$= \frac{7}{24} + \frac{9}{24}$

$= \frac{1}{2}$

$P(X \geq 3) = 0.6667$

⑦ $n = 20, p = 0.08, q = 1, P = 0.92$

⑧ $P(X=2)$

$= \binom{20}{2} (0.08)^2 (0.92)^{18}$

$= 0.2711$

⑨ $P(X \geq 3) = 1 - P(X \leq 2)$
 $P(X \geq 3) = 1 - [P(X=0) + P(X=1) + P(X=2)]$

$= 0.2121$

⑩ $P = (20)(0.08) = 1.6$

$\sigma = \sqrt{np(1-p)}$

$= \sqrt{20(0.08)(0.92)}$

$= \sqrt{1.472}$

$= 1.2133$

⑪ $\lambda = 4.5$

⑫ $P(X=6) = \frac{e^{-4.5} (4.5)^6}{6!}$

$= 0.1281$

⑬ $X \leq 2$

$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$= \frac{e^{-4.5} (4.5)^0}{0!} + \frac{e^{-4.5} (4.5)^1}{1!} + \frac{e^{-4.5} (4.5)^2}{2!}$

$= 0.1736$

⑭ $4.5, 2$

$\lambda = (4.5)(2)$

$= 9$

$P(X > 10) = 1 - P(X \leq 10)$

$= 1 - \sum_{k=0}^{10} \frac{e^{-9} 9^k}{k!}$

$= 0.2940$

⑮ $P(X > 135)$

⑯ $Z = \frac{135 - 120}{15}$

$Z = 1$
 $= P(Z > 1) = 1 - P(Z \leq 1)$

$= 1 - 0.8413$

$= 0.1587$

⑰ $P(100 < X < 130)$

$Z = \frac{100 - 120}{15} = -1.3333$

$= \frac{130 - 120}{15} = 0.6667$

$P(100 < X < 130) = P(-1.3333 < Z < 0.6667)$

0.6563

⑱ $P(X \leq x) = 0.95$

$Z = 1.6449$

$X = 120 + (1.6449)(15)$

$= 144.6728 \text{ ms}$